Light

GCSE Computer Science — CPU Architecture (Set 2)

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A program needs to check whether a user's password matches the stored password.
Which component of the CPU performs this comparison?

- ○ **A.** The Memory Data Register (MDR)
- ○ **B.** The Program Counter (PC)
- ○ **C.** The Control Unit (CU)
- ○ **D.** The Arithmetic Logic Unit (ALU)

Q2 (1 mark)

Before data can be read from RAM, the CPU must specify which memory location to access.

Which register holds the address of the memory location being accessed?

- ○ **A.** The Memory Address Register (MAR)
- ○ **B.** The Current Instruction Register (CIR)
- ○ **C.** The Accumulator (ACC)
- ○ **D.** The Program Counter (PC)

Q3 (1 mark)

A computer is saving a document. The data must be transferred from the CPU to RAM for storage.
Which bus carries this data between the CPU and RAM?

- ○ **A.** The control bus
- ○ **B.** The address bus
- ○ **C.** The system bus
- ○ **D.** The data bus

Q4 (1 mark)

During program execution, different CPU components must work together at the correct times. One component sends signals telling other components what to do.
Which component sends these control signals?

- ○ **A.** The Control Unit (CU)
- ○ **B.** The Accumulator (ACC)
- ○ **C.** The Clock

- ○ **D.** The Arithmetic Logic Unit (ALU)

Q5 (1 mark)

The Accumulator (ACC) is a special register in the CPU.
What is the main purpose of the Accumulator?

- ○ **A.** To store the address of the next instruction to be fetched
- ○ **B.** To hold the instruction currently being decoded
- ○ **C.** To send addresses to RAM via the address bus
- ○ **D.** To store results from ALU calculations

Q6 (1 mark)

The CPU processes instructions using a continuous cycle. In the final stage of each cycle, the instruction is actually carried out.
What is this final stage called?

- ○ **A.** Decode
- ○ **B.** Fetch
- ○ **C.** Store
- ○ **D.** Execute

Q7 (1 mark)

Von Neumann architecture uses the 'stored program concept'.
What does this mean?

- ○ **A.** Programs and data are both stored in the same main memory (RAM)
- ○ **B.** Programs are stored on external USB drives and loaded when needed
- ○ **C.** Programs are permanently stored in the CPU and cannot be changed
- ○ **D.** Each program must be stored on a separate hard drive

Q8 (1 mark)

The address bus connects the CPU to RAM.
In which direction does data flow on the address bus?

- ○ **A.** From CPU to RAM only (unidirectional)
- ○ **B.** From RAM to CPU only (unidirectional)
- ○ **C.** Both directions (bidirectional)
- ○ **D.** The address bus does not carry any data

Q9 (1 mark)

The CPU needs to tell RAM whether it wants to read data or write data.
This read/write signal must travel from the CPU to memory.
Which bus carries this type of signal?

- ○ **A.** The data bus
- ○ **B.** The memory bus
- ○ **C.** The address bus

- ○ **D.** The control bus

Q10 (1 mark)

During the fetch-decode-execute cycle, an instruction is copied from the MDR to the CIR.
What happens to the instruction once it is in the CIR?

- ○ **A.** It is immediately deleted to make room for the next instruction
- ○ **B.** It is sent back to RAM for permanent storage
- ○ **C.** It is decoded by the Control Unit to determine what operation to perform
- ○ **D.** It is converted from binary to decimal for the ALU

Q11 (1 mark)

After an instruction has been fetched from memory, the CPU must work out what the instruction means before it can be carried out.
What is this stage of the cycle called?

- ○ **A.** Execute
- ○ **B.** Interpret
- ○ **C.** Fetch
- ○ **D.** Decode

Q12 (1 mark)

Instructions in a program are normally executed one after another in sequence. After fetching each instruction, the CPU automatically

prepares to fetch the next one.
What happens to the Program Counter to enable this?

- ○ **A.** It is reset to zero
- ○ **B.** It is doubled
- ○ **C.** It is incremented
- ○ **D.** It is copied to the MAR

Q13 (2 marks)

During the fetch stage of the FDE cycle, several operations occur in sequence.
Which sequence correctly describes the fetch stage?

- ○ **A.** PC → MAR → RAM sends instruction → MDR → CIR, then PC incremented
- ○ **B.** CIR → MDR → MAR → RAM → PC incremented
- ○ **C.** ALU → ACC → MDR → RAM → PC incremented
- ○ **D.** MAR → PC → CIR → MDR → RAM sends instruction

Q14 (2 marks)

Two computers have different bus specifications:
Computer A: 32-bit data bus
Computer B: 64-bit data bus
Both computers have identical clock speeds and need to transfer a 256-bit block of data from RAM to the CPU.
Which statement correctly compares how these computers will perform this transfer?

- ○ **A.** Computer B will transfer the data in 4 operations, while

Computer A will need 8 operations.

○ **B.** Both computers will transfer the data in a single operation because modern buses can expand to fit any data size.

○ **C.** Computer A will transfer the data in 4 operations, while Computer B will transfer it in 8 operations.

○ **D.** Computer A will be faster because narrower buses have less electrical interference.

Q15 (2 marks)

During one complete fetch-decode-execute cycle, several registers are used.
Which statement correctly describes the roles of the PC, MAR, MDR, and CIR during the fetch stage?

○ **A.** The CIR sends the address to the MAR, the MDR receives the instruction, the PC stores the result, and the cycle repeats.

○ **B.** The MDR holds the address throughout, the MAR receives data from RAM, the CIR stores it temporarily, and the PC counts completed cycles.

○ **C.** The MAR generates the next address, sends it to the PC, the CIR fetches from memory, and the MDR decodes the instruction.

○ **D.** The PC provides the address, which goes to the MAR, the instruction comes back via the MDR, and is then placed in the CIR for decoding.

Q16 (2 marks)

The control bus carries several different types of signals between the CPU and other components.
Which list correctly identifies signals carried by the control bus?

- ○ **A.** Memory addresses, data values, and instruction opcodes
- ○ **B.** Voltage levels, resistance values, and current measurements
- ○ **C.** Binary numbers, ASCII characters, and pixel colours
- ○ **D.** Read/write signals, clock signals, and interrupt requests

Q17 (2 marks)

An older computer system has a 24-bit address bus. The manufacturer wants to upgrade the RAM from 8 MB to 32 MB.
Which statement correctly explains whether this upgrade is possible?

- ○ **A.** The upgrade is possible because address bus width can be dynamically increased when more RAM is installed.
- ○ **B.** The upgrade will work but the extra RAM will only be usable by the operating system, not by applications.
- ○ **C.** The upgrade is not fully possible because a 24-bit address bus can only address up to 16 MB of memory.
- ○ **D.** The upgrade will work perfectly because address bus width has no effect on maximum RAM capacity.

Q18 (2 marks)

The ALU (Arithmetic Logic Unit) performs two main types of operations. Which statement correctly describes both types of ALU operations?

- ○ **A.** Fetching instructions from RAM and decoding them for execution
- ○ **B.** Managing the clock signal and generating interrupt requests
- ○ **C.** Storing data in registers and transferring data to RAM

○ **D.** Arithmetic operations (addition, subtraction) and logic operations (AND, OR, NOT, comparisons)

Q19 (2 marks)

Embedded systems are found in many everyday devices.
Which list contains only devices that typically use embedded systems?

○ **A.** Laptop computer, desktop PC, gaming console

○ **B.** Washing machine, microwave oven, car engine management system

○ **C.** Smartphone, tablet, smart TV

○ **D.** Web server, database server, cloud computing platform

Q20 (2 marks)

Von Neumann architecture defines the basic structure of most modern computers.
Which list correctly identifies all the main components specified in Von Neumann architecture?

○ **A.** CPU, GPU, RAM, hard drive, and monitor

○ **B.** CPU (with ALU, CU, and registers), main memory (RAM), buses, and input/output

○ **C.** Motherboard, power supply, cooling fan, and case

○ **D.** Keyboard, mouse, screen, and speakers

Q21 (2 marks)

The Program Counter (PC) is automatically incremented after each instruction is fetched.
Why is this automatic increment necessary?

- ○ **A.** To count how many instructions have been executed for performance monitoring
- ○ **B.** To ensure the CPU fetches instructions in sequential order from consecutive memory addresses
- ○ **C.** To increase the clock speed as more instructions are processed
- ○ **D.** To make space in RAM for new instructions to be loaded

Q22 (2 marks)

The control bus is described as bidirectional because signals can travel in both directions.
Which example correctly demonstrates signals travelling in opposite directions on the control bus?

- ○ **A.** Data travelling from CPU to RAM, and data travelling from RAM to CPU.
- ○ **B.** Memory addresses travelling from CPU to RAM, and different addresses travelling from RAM to CPU.
- ○ **C.** Read/write signals travelling from CPU to RAM, and interrupt requests travelling from devices to CPU.
- ○ **D.** Clock signals travelling from CPU to devices, and faster clock signals travelling from devices to CPU.

Q23 (2 marks)

A student asks why all CPU operations need to be synchronised to a clock signal.
Which statement provides the best explanation?

○ **A.** The clock signal provides power to each component, and without it the CPU would have no electricity.

○ **B.** The clock ensures all components move to the next step together, preventing data from arriving before a component is ready to receive it.

○ **C.** The clock determines which programs are allowed to run and blocks unauthorised software.

○ **D.** The clock signal encrypts data as it travels between components to maintain security.

Q24 (3 marks)

A computer system has an 18-bit address bus.
Which calculation correctly determines the maximum amount of memory this system can address?

○ **A.** 18 bytes, because each bit represents one byte of addressable memory

○ **B.** 18 KB, because address bus width in bits equals memory in kilobytes

○ **C.** 262,144 bytes (256 KiB), because 2^18 = 262,144 addressable locations

○ **D.** 144 bytes, because 18 × 8 = 144 bits of addressable space

Q25 (3 marks)

A CPU runs at 2.5 GHz. A simple program requires 500 million clock cycles to complete.
How long will this program take to execute?

○ **A.** 0.2 seconds, because 500 million ÷ 2.5 billion = 0.2

○ **B.** 1.25 seconds, because 2.5 billion ÷ 500 million = 1.25

○ **C.** 200 seconds, because 500 million × 2.5 = 1250 million, then ÷ 1000

○ **D.** 5 seconds, because 500 million ÷ 100 million (per 1 GHz) = 5

Q26 (3 marks)

Two computer systems need to transfer 128 bytes from RAM to the CPU:
• System A has a 32-bit data bus
• System B has a 16-bit data bus
Assuming each bus transfer takes the same amount of time, which analysis is correct?

○ **A.** System A requires 32 transfers, System B requires 16 transfers; System B is faster

○ **B.** System A requires 32 transfers, System B requires 64 transfers; System A is twice as fast

○ **C.** System A requires 4 transfers, System B requires 8 transfers; System A is twice as fast

○ **D.** Both systems require 128 transfers because each byte needs its own transfer

Q27 (3 marks)

A CPU is about to fetch an instruction. The current state is:
• PC contains address 2048
• MAR contains address 1000
• MDR contains the value 42
• CIR contains a previous instruction
After the fetch stage completes, which register contents will have changed?

- ○ **A.** Only the PC will change (incremented to 2049)
- ○ **B.** The MAR (now 2048), MDR (new instruction from address 2048), CIR (copy of new instruction), and PC (incremented to 2049) will all change
- ○ **C.** Only the MDR and CIR will change; the PC and MAR stay the same
- ○ **D.** All registers will contain the same values because fetch does not modify registers

Q28 (3 marks)

An ADD instruction needs to add a value from memory address 500 to the current value in the Accumulator.
Which sequence correctly describes what happens during the execute stage of this instruction?

- ○ **A.** The value 500 is added directly to the ACC because the instruction contains the value
- ○ **B.** MAR ← 500, RAM sends value at address 500 to MDR, ALU adds MDR value to ACC, result stored in ACC
- ○ **C.** The CIR sends 500 to the ALU, which adds it to whatever is in RAM
- ○ **D.** PC ← 500, causing the next instruction to be fetched from address 500

Q29 (3 marks)

A student states: 'If we double the width of the data bus, data transfers will always be exactly twice as fast.'
Which response correctly evaluates this statement?

- ○ **A.** Correct — doubling bus width always doubles transfer speed because twice as many bits move per transfer
- ○ **B.** Incorrect — while each transfer moves more data, the overall speed also depends on RAM speed, bus frequency, and whether data sizes align with the bus width
- ○ **C.** Correct — data bus width is the only factor affecting transfer speed
- ○ **D.** Incorrect — doubling the data bus width actually slows transfers because more wires cause more interference

Q30 (3 marks)

A company is choosing a processor for a new smart thermostat that monitors temperature and controls heating.
Which factors are most important when selecting a processor for this embedded application?

- ○ **A.** Maximum clock speed and largest possible RAM to ensure fast response times
- ○ **B.** Low power consumption, reliability, small physical size, and sufficient processing power for the specific task
- ○ **C.** Ability to run multiple operating systems and support for high-resolution graphics
- ○ **D.** Compatibility with gaming software and support for virtual reality applications

Total: **55** marks • Time: **30 min**

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